29 June 1972
SUPERSEDING
MIL-B-43282A
2 November 1967

MILITARY SPECIFICATION

BALANCES, TORSION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers torsion balances (see 6.1).

1.2 Classification.- Torsion balances shall be of the following types and sizes as specified (see 6.2).

Type I - Pan or dish load receivers.

Size 1 - 4500 gram capacity, 0.4 gram sensitivity requirement (SR).
Size 2 - 2000 gram capacity, 0.25 gram sensitivity requirement (SR).
Size 3 - 4500 gram capacity, 0.35 gram sensitivity requirement (SR).

Type II - Plate load receiver (Cream Test Scale).

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

AAA-W-200 - Weight Sets, Balances
PPP-B-1122 - Balances, Scale and Accessories, Packaging Packing Of

FSC 6670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook 44 - Specifications, Tolerances, and Other Technical Requirements for Commercial Weighing and Measuring Devices, Adopted by the National Conference on Weights and Measures.

Handbook 77, Volume III,
Circular No. 3 - Precision Measurements and Calibration Optics, Metrology and Radiation, - Design and Test of Standards of Mass.

(Application for copies of Handbook 44 and Handbook 77 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C.).

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.4).

3.2 Standard product.- The balances delivered under this specification shall be the manufacturer's standard commercial product, except for changes necessary to comply with this specification.

* 3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the supplier for torsion balances provided the completed items comply with all the provisions of the specification.

* 3.4 Design and construction.- The type I and type II balances shall be equal arm scales with the characteristics specified in table I and hereinafter.

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The applied load shall create a torsional effect on the metal components of the balances. Counterbalance of this torsional effect shall give the weight of the applied load. The balance shall have adjustable poise weights to position the center of gravity of the balances at the center of rotation of the load receivers. The balances shall be inclosed and shall have three metal trusses. The center truss shall be attached to a base. The counterbalance shall utilize either a combination of calibrated spiral spring mechanism, weighbeam or removable weights. The balance shall meet the applicable retroactive and nonretroactive specifications specified in the National Bureau of Standards Handbook 44.

TABLE I.- Characteristics

	Type I Size 1	Type I Size 2	Type I Size 3	Type II
Capacity, grams	4500	2000	4000	240
Sensitivity requirement, gram 1/	0.5	0.25	0.35	0.032
Weighbeam				
Graduation, grams	0-500	0-100	0-100	
Intervals, grams	100	10	1.0	
Dial weighing mechanisms				
Graduation, gram	0-100	0-10		
Interval, gram	1	0.1		
Load receivers				
Diameter, (outside), minimum, inches	9	6-1/4	9	1-3/8 2/
Depth, minimum, inch	3/4	7/8	3/4	

1/ Sensitivity requirement is the amount of weight required to change the rest position of the indicator one scale division.
2/ Inside diameter of bottle support frame.

3.4.1 Type I.- The type I balance shall be an equal arm scale with a weighbeam. The calibrated dial mechanism and the weighbeam shall have the range

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of graduations and the intervals specified in table I. The balance shall have two interchangeable pan or dish load receivers.

3.4.2 Type II.- The type II balance shall be an equal arm scale for weighing 9 and 18 gram capacity Babcock bottles. The balance shall be designed to weigh the tare of the bottle and level the balance at zero point in one operation. The balance shall have a tare dial that will allow compensation for the weight of Babcock bottles. The balance shall have two plate load receivers. One load receiver shall have a support frame for one Babcock bottle. The overall dimensions of the balance shall be 11-3/4 inches plus or minus 1/4 inch long, 5-3/4 inches plus or minus 1/4 inch wide, and 7-3/4 inches plus or minus 1/4 inch high.

3.4.3 Components.-

3.4.3.1 Base.- The base on the type I and II balance shall support the balance and shall have leveling screws mounted on the bottom. The arrest mechanism shall be operated by a knob attached to the base.

3.4.3.2 Load receivers.- The load receivers shall be corrosion-resisting metal or nickel plated or chrome plated metal.

3.4.3.3 Support frame.- The bottle support frame for the type II balance shall be constructed of the same metal as the load receivers. The support frame shall be securely attached to the load receiver.

3.4.3.4 Enclosure.- The balances shall be enclosed in glass, metal or both glass and metal. The enclosure shall not include the weighbeam or load receivers (type I) and shall be removable to provide access for repairs and adjustment to the balance.

3.4.3.5 Torsion bands.- The torsion bands shall be constructed of corrosion-resisting steel in accordance with manufacturer's standard commercial practice.

* 3.5 Weights.-

3.5.1 The weights shall be contained in a wooden or plastic block with lid in accordance with manufacturer's standard commercial practice.

3.5.2 Type I.- Unless otherwise specified (see 6.2), the type I balance shall be provided weights in accordance with AAA-W-200 except as follows:

a. Size 1 and 3 shall conform to type VI but shall consist of the following values. One each 2,000, 1,000, 500, 200, 50, 20, 5, 2 and 1 gram and 2 each 100 and 10 gram.

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b. Size 2 shall conform to type VI but shall consist of 13 weights of the following values: one each 1,000, 500, 200, 50, 20, 5 and 1 gram and 2 each 100, 10 and 2 gram.

3.5.3 Type II.- The type II balance shall have a 9 and 18 gram weight. The weight shall conform to the requirements for class C weights as specified in Handbook 77, Volume III, circular No. 3 of the National Bureau of Standards.

3.6 Performance.-

3.6.1 Sensitivity requirement.- The sensitivity requirement (SR) specified in table I shall be tested as specified in 4.4.1.

3.6.2 Acceptance tolerances.- The acceptance tolerance for the type I balance shall be as specified in National Bureau of Standards Handbook 44 for small capacity scales. The acceptance tolerance for the type II balance shall be as specified in National Bureau of Standards Handbook 44 for cream test scales. Acceptance tolerance shall be tested as specified in 4.4.2.

3.7 Finish.- The type I balance shall be finished in accordance with the manufacturer's standard commercial practice. The type II balance shall have a white or gray baked enamel finish.

* 3.8 Marking.- The manufacturer's name, trade name, or trademark of such known character as to be easily identifiable shall appear on each balance. The above markings shall be in accordance with MIL-STD-130.

3.9 Repair and maintenance.- Provisions shall be made to allow for ready adjustment, service and replacement of minor components without disturbing or disassembly of any major components. In addition, there shall be free access to inspect, service and adjust any component as provided for by this specification, or as indicated by the component or equipment manufacturer, without requiring disassembly of any other component on the unit.

3.10 Workmanship.- The finished balance shall be free of sharp burrs, slivers, splinters or dents. The finish shall be free of sags, orange peel, wrinkles and areas of no film.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a first article is required, it shall be examined in accordance with the defects in table II, specified dimensions and tested as specified in 4.4.1 and 4.4.2. The presence of any visual defects, any dimensions not as specified, or failure to pass any test shall be cause for rejection of the first article.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- The lot shall be the number of balances of one type and size offered for acceptance at one time. The sample unit for this inspection shall be one completely fabricated balance.

4.3.2.1 Visual examination.- Examination of the end item shall be in accordance with the defects listed in table II. The inspection level shall be level II and the acceptable quality level shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished as required, runs, sags, wrinkles, areas of no film		X
	Color of type II scale not as specified		X
Design	Any characteristics not in accordance with the specified requirements	X	

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (applicable to all components and assemblies).	Part missing or not specified type	X	
	Any component not readily accessible for servicing where required	X	
	Component not properly assembled or secured	X	
	Fractured, malformed or otherwise impaired	X	
	Sharp burr, sliver, splinter or dents	X	
Base	Leveling screws missing	X	
	No arrest knob mechanism attached to base	X	
Load receivers	Support frame missing (when applicable)	X	
	Not plates or pans or dish (as applicable)		X
Support frame	Not interchangeable (when applicable)	X	
	Not secured to load receiver	X	
Marking identification	Missing, incomplete, not legible, not as specified		X

4.3.2.2 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-3 and the acceptable quality level (AQL) shall be 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- The end item shall be tested for the applicable characteristics specified in 4.4.1 and 4.4.2. The inspection level shall be S-2 and the AQL shall be 2.5 defects, expressed in terms of defects per hundred units.

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4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing and marking comply with the requirements of PPP-B-1122.

4.4 Tests.-

4.4.1 Sensitivity requirement.-

4.4.1.1 Type I.- The type I balance shall be tested for sensitivity requirement (SR) loads equal to 0, 1/8, 1/2 and full capacity of the balance. Weights equal to each of these loads except at zero reading shall be applied respectively on the load receivers. A 1-gram weight shall then be applied to a load receiver and the sensitivity requirement (SR) calculated to determine compliance with the requirements specified in 3.6.1. Any evidence that the sensitivity requirements are not within the calculated tolerance shall constitute failure of this test.

4.4.1.2 Type II.- The sensitivity requirement (SR) of the type II balance shall be determined at a load equal to 2 avoirdupois ounces. A weight equal to this load shall be applied on the load receivers and a 1-grain weight applied to a load receiver and sensitivity requirement (SR) calculated to determine compliance with the requirements specified in 3.6.1. Any evidence that the sensitivity requirements are not within calculated tolerance shall constitute failure of this test.

4.4.2 Acceptance tolerance.- The type I and type II balances shall be tested for the acceptance tolerance in accordance with the National Bureau of Standards Handbook 44, for small capacity and cream test scales respectively to determine compliance with the requirements specified in 3.6.2. Any evidence that the acceptance tolerance is not in accordance with NBS Handbook 44 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking for level A, B, or C shall be in accordance with PPP-B-1122 (see 6.2).

6. NOTES

6.1 Intended use.- The type I balance covered by this specification is intended primarily for the weighing of dry powders and other objects within the capacities specified. The type II balance is used for weighing the butter fat content of milk.

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* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size, required (see 1.2).
(c) When first article is required (see 3.1, 4.2 and 6.4).
(d) Weight(s) required if other than specified (see 3.5.2).
(e) Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

6.2.1 Any requirement for equipment manuals for the item covered by specification should be included in DD Form 1423 Contract Requirements List cited in contract.

6.3 The margin of this specification has been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) were made to previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.4 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of Armed Services Procurement Regulations ASPR 7-104.55. The first article should be a preproduction sample, initial production item, or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodians:

Army - GL
Air Force - 82

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 6670-0098

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Balances, Torsion MIL-B-43282B

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

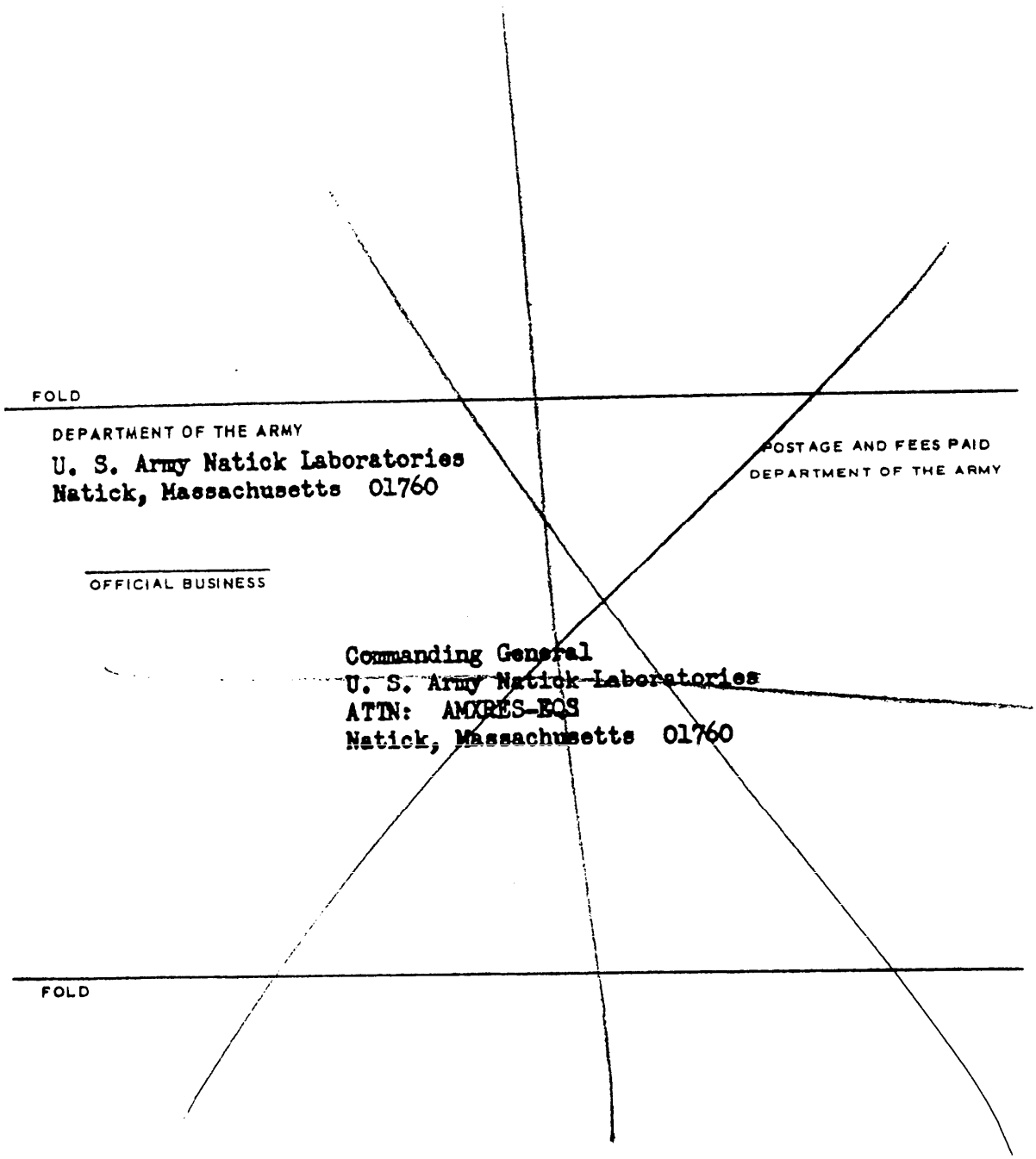
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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